

The Exponential Wall

Qubits	Amplitudes	State size	
20	10^6	16 MB	fits in cache
30	10^9	16 GB	fits in RAM
38	3×10^{11}	2 TB	needs disk cluster $2 \times \text{qubits}$
45	4×10^{13}	256 TB	large cluster $= 2 \times \text{state}$
50	10^{15}	16 PB	supercomputer



Once simulation spills to disk and runs for **hours or days**...

**What happens when a
node crashes mid-run?**

You lose all progress and start over.

This thesis: make large-scale state-vector
simulation **restartable** and **crash-safe**